

Readout-compatible CA1 plasticity supports cue-dependent reconfiguration and degraded-cue recall in a minimal hippocampal model

Abstract

Rapid memory formation requires retrieved content to remain interpretable by downstream circuits. We examine this requirement, termed *readout compatibility*, in a minimal hippocampal model. Entorhinal cortex (EC) input generates a CA3-like retrieval key and a pretrained CA1 target; plastic CA3–CA1 synapses associate the two, and a fixed CA1–EC decoder reconstructs the input. Across 20 paired initializations, permuting the target’s CA1 coordinates impaired one-shot reconstruction under both an instructive-driven rule (mean output–target cosine: 0.880 aligned, 0.247 permuted) and a bounded error-driven rule (0.872, 0.178). Correspondingly permuting the decoder restored performance, isolating a coordinate mismatch rather than loss of information from the target. On a circular track, exchanging cue identities produced reproducible changes in recalled CA1 tuning and alternating population states associated with cue assignment. Both rules generated heterogeneous spatially stable and cue-modulated responses. During frozen recall, selective lateral or medial EC degradation preferentially impaired cue or position readout, respectively; a common-key control that eliminated item-specific CA3 activity failed even with intact inputs. Separate parameter searches identified distinct operating regimes, and held-out sensitivity and load analyses revealed parameter dependence and interference across up to 90 ordered cue pairs. These simulations show how rapid associative learning can express retrieved content in coordinates compatible with an established readout. They demonstrate cue-dependent reconfiguration and degraded-input robustness in a feedforward architecture, not recurrent CA3 pattern completion, a biological mechanism for maintaining compatibility, or long-term memory stability.

1 Introduction

Episodic memory requires a circuit to distinguish related experiences, retrieve stored content from a partial cue, and express that content in a form that other circuits can use. Hippocampal-indexing and complementary-learning-systems theories explain how rapid hippocampal storage can complement slower cortical learning [16, 23]. Within this framework, pattern separation limits interference and pattern completion supports retrieval. Here we focus on a complementary requirement: compatibility between a retrieved population state and the downstream mapping that interprets it. A representation may preserve information while changing the correspondence between neural activity and its readout. Successful retrieval therefore depends not only on what is stored, but also on the coordinates in which it is expressed.

The hippocampal circuit provides a useful setting in which to examine this requirement. Entorhinal cortex (EC) reaches CA1 through both the trisynaptic pathway, via dentate gyrus and CA3, and the

direct temporoammonic projection. Autoassociative theories assign recurrent CA3 circuitry a central role in content-addressable retrieval [15], consistent with impaired partial-cue recall after selective disruption of CA3 NMDA-receptor function [19]. CA3 and CA1 also exhibit distinct population responses to environmental changes [11]. Computational accounts already give CA1 and its direct entorhinal input roles beyond relaying CA3 output: in a complementary-learning-systems model within the hippocampus, the monosynaptic pathway learns statistical structure while the trisynaptic pathway supports memory for individual episodes [22]. More recently, scaffold-based models have separated content storage from the dynamics that generate stable retrieval states [2]. Our question is narrower than either account: given an established output mapping, how does the compatibility of a newly instructed CA1 state affect its subsequent reconstruction?

Rapid instructed plasticity provides a candidate computational motif. Behavioral-timescale synaptic plasticity (BTSP) can create a CA1 place field after a single dendritic plateau event and bidirectionally reshape existing fields [1, 17]. EC layer 3 activity can provide teaching-related input for learning-dependent CA1 reorganization [7]. Experiments and models link BTSP-like dynamics to trial-by-trial field shifts and stabilization of task-relevant spatial codes [14, 24], while computational work shows how BTSP-inspired rules can support rapid content-addressable storage [25]. Recent evidence also extends the empirical motivation beyond spatial tasks: large calcium events consistent with BTSP accompany, and experimentally induced events can drive, rapid formation of odor responses in CA1 [4]. These findings motivate an EC-related instruction that modifies CA3–CA1 synapses. They do not establish that a biological instructive signal specifies the complete population target used in our model.

The proposed constraint also relates to work on neural population geometry and readout stability. In motor-cortical brain–computer interfaces, perturbations within an existing neural manifold are learned more readily than perturbations requiring activity outside it [21]; short-term adaptation can also reassign an established activity repertoire to different outputs [6]. Stable latent dynamics can support cross-day behavioral decoding even when fixed decoders applied directly to recorded neurons deteriorate [5]. Such latent stability should not be equated with stability of a neuron-by-neuron readout. Conversely, models of adaptive readouts show how downstream plasticity can compensate for an evolving population code [20]. We use *readout compatibility* as a functional criterion: a recalled CA1 state is compatible to the extent that an established decoder reconstructs its intended content. This criterion neither requires every neuron to retain its tuning nor identifies compatibility with membership of a particular neural manifold.

To make this criterion explicit, we pretrain an EC–CA1–EC autoencoder and then freeze its encoder and decoder. Plastic CA3–CA1 synapses subsequently associate a feedforward CA3-like retrieval key with the encoder-defined CA1 target. The pretrained basis abstracts structure accumulated before the associative learning assay; it is not a proposed biological backpropagation mechanism. Prior theoretical work motivates using established structure to support new associations [2, 22], and pretraining offers an engineering analogue of feature reuse [26]. Freezing the decoder prevents downstream adaptation from compensating for changes in the instructed coordinates. The model thus tests the consequences of compatibility under controlled conditions, rather than explaining how biological circuits acquire or maintain it.

We compare an instructive-driven update, which depends on the target but not the recalled CA1 state, with a bounded error-driven update that corrects the target–recall discrepancy. First, a coordinate-permutation assay separates information preserved in an instructed vector from its accessibility to a fixed decoder. Second, a cue-guided track assay tests context-dependent CA1 reconfiguration.

Third, selective input degradation tests reconstruction when spatial or cue information is reduced. For these manipulations, EC is partitioned into MEC-like spatial and LEC-like cue inputs. This separation is experimental bookkeeping, not an anatomical claim: both medial and lateral EC contain mixtures of spatial and nonspatial signals [3, 4, 9, 10]. Finally, rule-specific parameter searches, held-out sensitivity analyses, and an ordered-cue-pair load assay assess how the results depend on the operating regime. Together, these tests evaluate a representational constraint on rapid associative learning without attributing recurrent dynamics or long-term stability to this minimal circuit.

2 Results

2.1 A minimal retrieval-key and instructed-CA1 architecture

The model contains four functional components (Fig. 1A). Entorhinal input, denoted x_{EC} , projects through a fixed sparse map to a CA3-like population that provides a retrieval key. A separately pretrained EC-CA1 encoder transforms the same input into an instructed CA1 state (IS). Plastic CA3-CA1 weights associate the retrieval key with the IS, whereas the corresponding pretrained CA1-EC decoder remains fixed. During recall, the EC input regenerates a key, the learned association reconstructs a CA1 state, and the decoder maps that state back to EC output. This architecture separates the selection of associative content from its expression in a pre-existing output basis.

We first evaluated two plasticity rules under matched inputs and a shared reference parameter configuration (Fig. 1B). The instructive-driven rule writes an outer-product association between the instructed CA1 state and the CA3-like key without considering the current recalled state. The bounded error-driven rule instead potentiates CA3-CA1 synapses for underexpressed CA1 components and depresses them for overexpressed components. Repeated presentations therefore rewrite the instructed association under the instructive-driven rule but correct a residual under the error-driven rule. Both are abstract models of rapid instructed plasticity; neither implements the temporal eligibility trace or dendritic plateau dynamics of biological BTSP.

Track inputs combined a MEC-like spatial component with a LEC-like cue component (Fig. 1D). Two cue identities occurred at fixed positions and could exchange locations across blocks. For degradation tests, the model first stored intact laps; plasticity was then frozen while a fixed fraction of either the MEC-like or LEC-like input units was set to zero (Fig. 1E).

2.2 Readout compatibility constrains rapid associative storage

We first isolated the relationship between an instructed CA1 code and its fixed output decoder (Fig. 2A). Twenty independently initialized autoencoders were pretrained on structured MEC+LEC patterns and then frozen. Pattern order was shuffled, so this assay retained the marginal spatial-cue structure without presenting a traversed track sequence. For each initialization, 28 held-out patterns were stored once under five paired conditions. In the aligned condition, the IS remained in the coordinates learned by the decoder. In the fixed-permutation condition, the same non-identity permutation was applied to every IS while the decoder was left unchanged. In the matched-decoder condition, the corresponding decoder columns were permuted together with the IS. Random-content and no-plasticity conditions controlled for storing an incorrect association and for leaving the

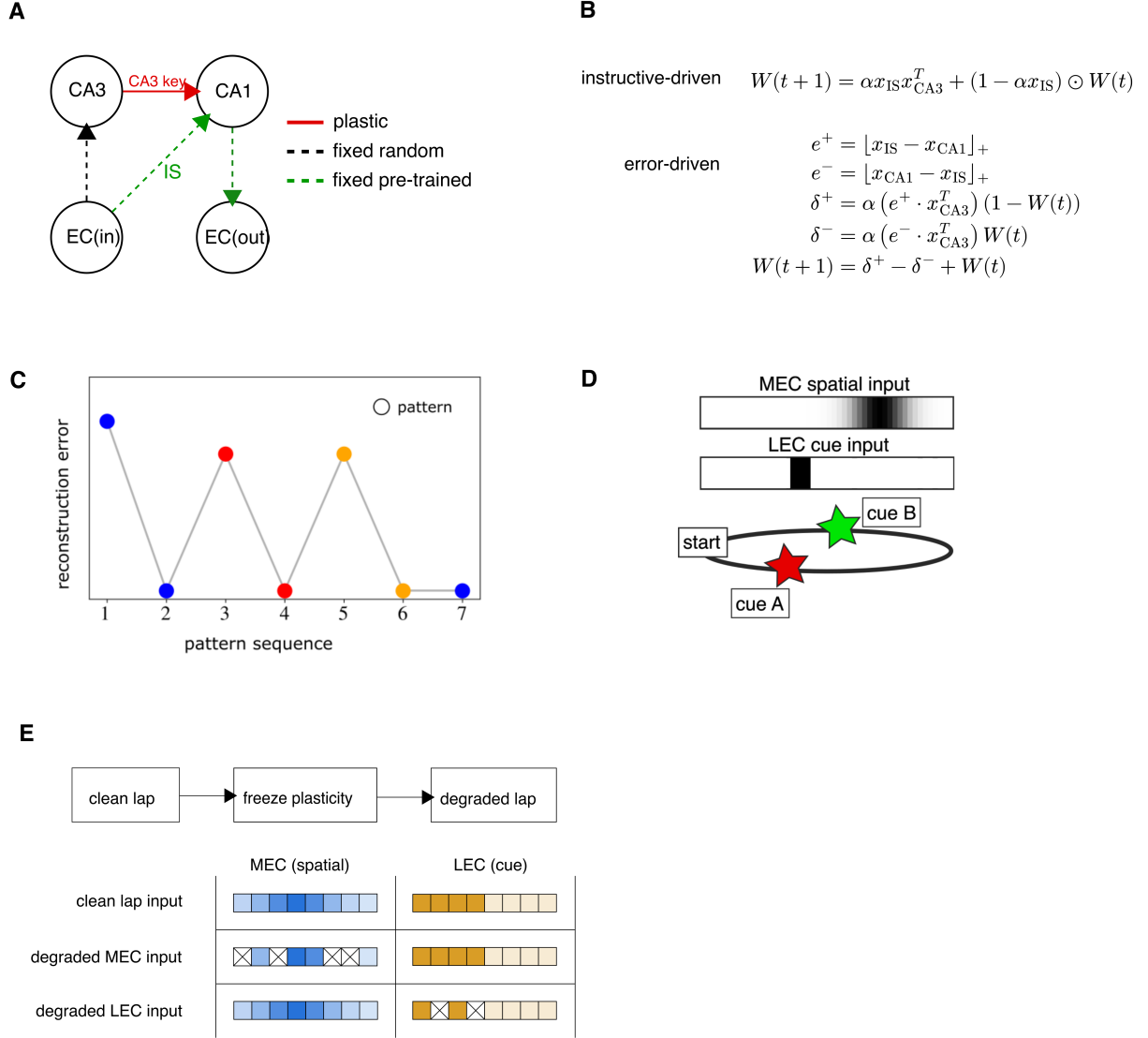


Figure 1: Model architecture and simulation protocol. (A) EC input produces a fixed sparse CA3-like retrieval key x_{CA3} and, through a pretrained encoder, an instructed CA1 state x_{IS} . Plastic CA3–CA1 weights store their association; a fixed pretrained CA1–EC decoder reads out recall. (B) Instructive-driven and bounded error-driven update rules. (C) Conceptual reconstruction-error trajectories over first and repeated presentations; colors denote pattern identity. This panel is a schematic rather than a measured result. (D) Circular-track input composed of MEC-like spatial activity and LEC-like cue activity. (E) Selective degradation protocol. Intact laps are stored before plasticity is frozen; a fraction of MEC-like or LEC-like units is then zeroed using a mask held fixed within a probe lap and resampled across probes.

memory unwritten, respectively.

Under the instructive-driven rule, aligned instructions produced a mean output–target cosine of 0.880 (95% confidence interval [CI], 0.870–0.891), whereas a fixed coordinate permutation reduced it to 0.247 (0.207–0.286). The paired aligned-minus-permuted difference was 0.634 (0.592–0.676). The same coordinate dependence appeared under the error-driven rule: aligned and permuted means were 0.872 (0.858–0.887) and 0.178 (0.147–0.209), respectively, giving a paired difference of 0.694 (0.658–0.731). Permuting the decoder together with the IS restored the aligned mean exactly for each rule. Random-content recall remained lower (0.355 for the instructive-driven rule and 0.315 for the error-driven rule), as did the shared no-plasticity control (0.114). The matched-decoder restoration is an algebraic coordinate control: the permutation preserves the IS values but changes their meaning to the unchanged decoder. Reconstruction therefore depends on compatibility between the instructed CA1 coordinates and the fixed output basis, not on information content alone. Under this reference configuration, the two rules reached similar aligned one-shot recall despite using different update signals: the instructive-driven rule performs a target-gated associative update, whereas the error-driven rule makes a bounded residual correction.

2.3 Cue exchanges induce reproducible CA1 reconfiguration

We next exposed the model to 40 track laps in which two cue identities exchanged positions every 10 laps (Fig. 2B). A paired no-swap control used the same root seeds, pretrained encoder and decoder, MEC-like trajectories, lap count, plasticity parameters, and exact EC–CA3 projection, but retained the initial cue assignment throughout. After every training lap, plasticity was paused and the model was probed in the context scheduled for that lap.

For the instructive-driven rule, mean unit-wise tuning similarity between consecutive probes fell to 0.365 at cue exchanges (95% CI, 0.348–0.382), compared with 0.845 at the same lap boundaries in the no-swap control (0.832–0.857). The paired swap-minus-control difference was -0.479 (-0.498 to -0.461). Under the error-driven rule, exchange-boundary similarity was 0.451 (0.436–0.466), compared with 0.827 in the matched control (0.817–0.838), for a paired difference of -0.376 (-0.396 to -0.356). At non-exchange transitions in the alternating schedule, similarity remained high for both the instructive-driven rule (0.833, 0.823–0.843) and the error-driven rule (0.821, 0.814–0.828). The abrupt tuning changes were therefore associated with the altered cue sequence rather than with continued learning alone.

In a separate realization of the same protocol, we aligned the three exchange events to their first post-exchange lap (Fig. 2E). Under the instructive-driven rule, consecutive-lap similarity at the exchange was 0.375 (95% CI, 0.351–0.398), versus 0.851 (0.839–0.863) at matched no-swap boundaries; the paired difference was -0.477 (-0.499 to -0.454). Under the error-driven rule, the corresponding values were 0.462 (0.443–0.480) and 0.833 (0.822–0.845), with a paired difference of -0.372 (-0.391 to -0.352). By the following lap, similarity recovered to 0.769 (0.756–0.783) and 0.819 (0.806–0.832), respectively. Thus the response was concentrated at the cue exchange rather than developing as nonspecific drift over the subsequent block.

Because the probe context also changes at an exchange, this discontinuity does not by itself distinguish synaptic updating from the immediate response to a different retrieval cue.

The lap-by-lap population representational-similarity matrix provided a complementary view (Fig. 2F). Within-block maps remained similar, exchanges separated the sequence into distinct

blocks, and nonadjacent blocks carrying the same cue assignment resembled one another. The model therefore alternated between cue-dependent population states rather than accumulating a wholly novel map after every block. Retrospective comparison with the final map of each new block further showed that the first post-exchange representation was already closer to the eventual adapted map than to the pre-exchange map, followed by smaller within-block changes (Supplementary Fig. S2A,B). Endpoint similarities were broadly distributed across CA1 units, demonstrating that the population effect was not produced by a single uniform response class (Supplementary Fig. S2C,D).

After learning, plasticity was frozen and both cue arrangements were probed. The representative error-driven population contained spatially stable fields, amplitude-modulated responses, and fields whose preferred position changed with cue identity (Fig. 2C). Across seeds, mean unit-wise, mean-centered spatial stability was 0.330 (95% CI, 0.297–0.363) under the instructive-driven rule and 0.464 (0.441–0.486) under the error-driven rule. Mean absolute cue modulation was 0.132 (0.130–0.135) and 0.124 (0.120–0.128), respectively. In both rules, the unit-level distributions were continuous rather than separated into discrete response classes (Fig. 2D), and strong cue modulation was concentrated among units with low or negative cross-context stability. These distributions are descriptive: units nested within a seed were not treated as independent replicates, and the analysis is not a formal classification of rate versus global remapping. Operationally, changing the cue arrangement changed both the EC-derived instruction and the CA3-like key, producing a mixture of stable and context-dependent recalled CA1 activity.

2.4 Selective EC degradation dissociates cue and spatial readout

We then asked whether the stored association remained usable when one EC input modality was selectively corrupted (Fig. 3). Each model stored 20 intact laps with a fixed cue arrangement. Plasticity was frozen, and independent masks requested removal of 0, 25, 50, 75, 90, or 100% of eligible units. Because each modality contained 25 units, the implemented fractions were 0, 24, 48, 76, 88, and 100%, respectively. Each fraction used 12 masks per seed and 20 paired root seeds; the numerical summary below focuses on 22-of-25 (88%) partial degradation. The principal control replaced the normal sparse EC–CA3 map with a dense common map in which every CA3-like unit received the same average EC input. This manipulation removes item-specific key structure and is therefore a degenerate-key, not a sparsity-matched, control.

Under LEC-like degradation, cue identity accuracy for the normal key declined from 1.000 to 0.871 at 88% dropout under the instructive-driven rule (95% CI at 88%, 0.803–0.938) and from 1.000 to 0.829 under the error-driven rule (0.766–0.893; Fig. 3A). The corresponding 88%-dropout changes were -0.129 (-0.197 to -0.062) and -0.171 (-0.234 to -0.107). MEC output–target cosine remained near its instructive-driven baseline (0.736 at baseline and 0.723 at 88% dropout; change -0.013 , -0.029 to 0.003), whereas it declined from 0.904 to 0.780 under the error-driven rule (change -0.125 , -0.136 to -0.113 ; Fig. 3B). Thus spatial output remained substantially readable under LEC-like corruption, although it was not strictly invariant under the error-driven rule.

MEC-like degradation produced the complementary loss of position information. Nearest-position accuracy declined from 0.163 to 0.070 under the instructive-driven rule (88%-dropout change -0.093 , 95% CI -0.106 to -0.079) and from 0.666 to 0.069 under the error-driven rule (change -0.597 , -0.636 to -0.557 ; chance = 0.02; Fig. 3C). Cue identity remained 1.000 for every seed and degradation level under both rules (Fig. 3D). The low baseline position accuracy under the instructive-driven rule is itself informative: repeated error-driven correction yielded a substantially

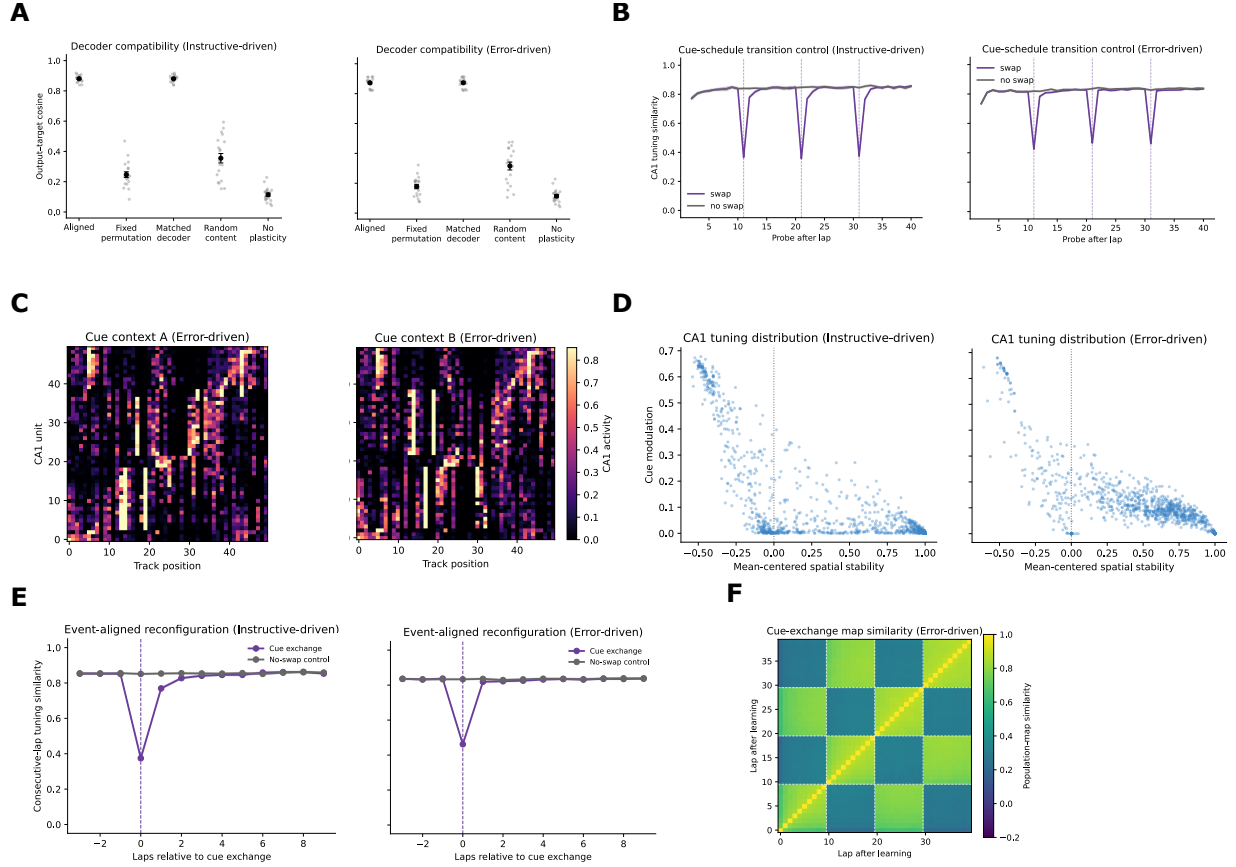


Figure 2: Readout compatibility and cue-dependent CA1 reconfiguration. (A) Output-target cosine for shuffled MEC+LEC samples in aligned, fixed-permutation, matched-decoder, random-content, and no-plasticity conditions for the instructive-driven rule (left) and error-driven rule (right). Grey points are seed means across 28 stored memories; black points and error bars show mean $\pm$ SEM across 20 seeds. (B) Mean unit-wise tuning similarity between consecutive scheduled-context probes for the instructive-driven rule (left) and error-driven rule (right). Purple denotes the cue-exchange schedule and grey the matched no-swap control; vertical dashed lines mark exchanges. Curves show mean $\pm$ SEM across 20 paired seeds. (C) Recalled CA1 activity on the same track under cue contexts A (left) and B (right) for the representative error-driven seed. Units have the same order and are sorted by preferred position across contexts; color limits are shared. (D) Unit-level relationship between mean-centered spatial stability and cue modulation for the instructive-driven rule (left) and error-driven rule (right), pooling 50 units from each of 20 seeds (1,000 points per rule). The vertical line marks zero stability. Panels C and D illustrate heterogeneous stable and cue-dependent responses but do not formally classify remapping type; confidence intervals in the text use seed-level summaries. (E) Event-aligned consecutive-lap tuning similarity for the instructive-driven rule (left) and error-driven rule (right) in the cue-exchange schedule and exact-wiring no-swap control. Lap zero is the first post-exchange lap; values were averaged across three exchanges within each seed and curves show mean $\pm$ SEM across 20 seeds. (F) Error-driven lap-by-lap population-map similarity, averaged across 20 seeds. Before comparison, each unit was mean-centered over track position and the full position-by-unit map was vectorized. Dashed lines mark cue-exchange boundaries. Panels B and E used identical EC-CA3 projections in the exchange and no-swap schedules for each rule; panel F shows the error-driven exchange schedule.

more discriminable spatial readout before degradation, whereas the instructive-driven rule did not. The two rules should therefore not be interpreted as starting from equivalent clean spatial performance.

The dense common-key control remained at chance for cue identity (0.5) and position (0.02) even without degradation, and produced low spatial-output cosine. Its failure establishes that a non-specific common key cannot support the associations, but does not isolate sparsity from key discriminability. Together, the normal-key results demonstrate degraded-cue robustness and pathway-selective failure within this architecture. They do not establish recurrent pattern completion: the CA3-like layer has no attractor dynamics, and recall deteriorates once the feedforward retrieval key loses discriminability. Supplementary Fig. S1 collects four seed-paired contrasts that summarize how the rule choice changes these effects.

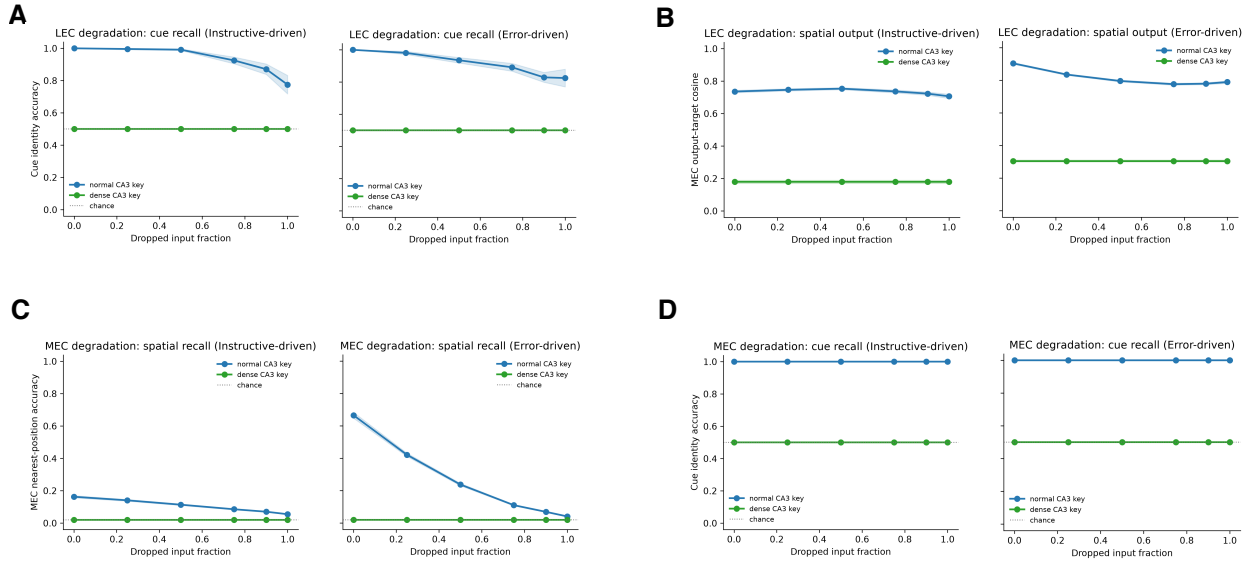


Figure 3: Selective EC degradation during frozen recall. Each pair of plots shows the instructive-driven rule on the left and the error-driven rule on the right, under the shared reference configuration. **(A)** Cue-identity accuracy during LEC-only degradation. **(B)** MEC output-target cosine during LEC-only degradation. **(C)** Nearest-position accuracy during MEC-only degradation. **(D)** Cue-identity accuracy during MEC-only degradation. Curves show mean $\pm$ SEM across 20 seeds after averaging 12 masks within each seed. Abscissae show requested fractions; whole-unit rounding gave effective fractions of 0, 24, 48, 76, 88, and 100%. Blue denotes the normal sparse key map; green denotes the dense common-key control, which eliminates item-specific key structure and is not sparsity matched. Dotted lines denote chance accuracy where applicable. Plasticity was disabled during every probe.

2.5 Parameter optimization identifies rule-specific operating regimes and load-dependent trade-offs

The matched reference configuration above isolates consequences of changing the update equation, but it need not be favorable to both rules. We therefore performed separate covariance matrix adaptation evolution strategy (CMA-ES) searches for the instructive-driven and error-driven rules on three fixed development seeds. Within each generated MEC+LEC lap, storage samples were shuffled before learning, preserving the input distribution while removing adjacent track states. The objective equally weighted clean recall, recall after 21-of-25 (84%) MEC-like degradation, and recall after 21-

of-25 (84%) LEC-like degradation (nominal requested fraction, 85%); these development evaluations were not treated as experimental replicates. To keep the representational basis common across the principal and optimization-informed analyses, these searches fixed $K_{CA1} = 5$ and both autoencoder gains at 25. The independently selected medial-temporal-lobe (MTL) model configurations used fan-in 36, $K_{CA3} = 6$, $\beta_{CA3} = 286.747$, $\beta_{CA1} = 9.160$, and $\alpha = 0.03366$ for the instructive-driven rule, versus fan-in 9, $K_{CA3} = 5$, $\beta_{CA3} = 54.400$, $\beta_{CA1} = 26.928$, and $\alpha = 0.08197$ for the error-driven rule. Their development-set composite scores were 0.714 and 0.818, respectively. Because both parameter vectors were selected on the same three seeds used to define the objective, these values describe optimization outcomes rather than unbiased estimates of generalization.

A one-parameter-at-a-time analysis evaluated multipliers from 0.1 to 9 around each selected vector while fixing all other parameters (Fig. 4A). Ten independent evaluation seeds, disjoint from the search seeds, supplied paired score differences and their across-seed standard deviations. The composite score changed smoothly near most selected values but was not uniformly insensitive. The largest mean decreases were produced by applying the largest K_{CA3} multiplier to instructive-driven plasticity (-0.231) and reducing β_{CA3} tenfold for error-driven plasticity (-0.477). The requested ninefold increase in K_{CA3} was clipped from 54 to the allowed maximum of 49. Small positive mean differences at a few tested points were at most 0.002 and were small relative to their across-seed variation. The analysis therefore supports a local operating region while revealing important parameter asymmetries; it is not a global robustness guarantee.

We next tested the selected configurations on 10 held-out seeds while increasing associative load from 1 to all 90 non-repeating ordered pairs drawn from 10 cue identities (Fig. 4B). Every pair was presented three times, and exact two-cue recall was measured from clean probes and after 12-of-25 (48%) LEC-unit dropout. Instructive-driven clean recall declined from 1.000 ± 0.000 at one pair to 0.210 ± 0.043 at 10 pairs and 0.138 ± 0.016 at 90 pairs (mean $\pm$ SEM); corresponding degraded recall was 1.000 ± 0.000 , 0.035 ± 0.012 , and 0.051 ± 0.006 . Error-driven clean recall was 1.000 ± 0.000 , 0.390 ± 0.046 , and 0.151 ± 0.022 at the same loads, with degraded values of 0.775 ± 0.095 , 0.098 ± 0.024 , and 0.046 ± 0.003 . Thus both selected regimes exhibited interference as load increased, with an error-driven advantage at intermediate clean load but convergence toward low exact-pair recall at the largest load. These curves are empirical load profiles for this cue vocabulary, training schedule, and model size rather than universal capacity estimates.

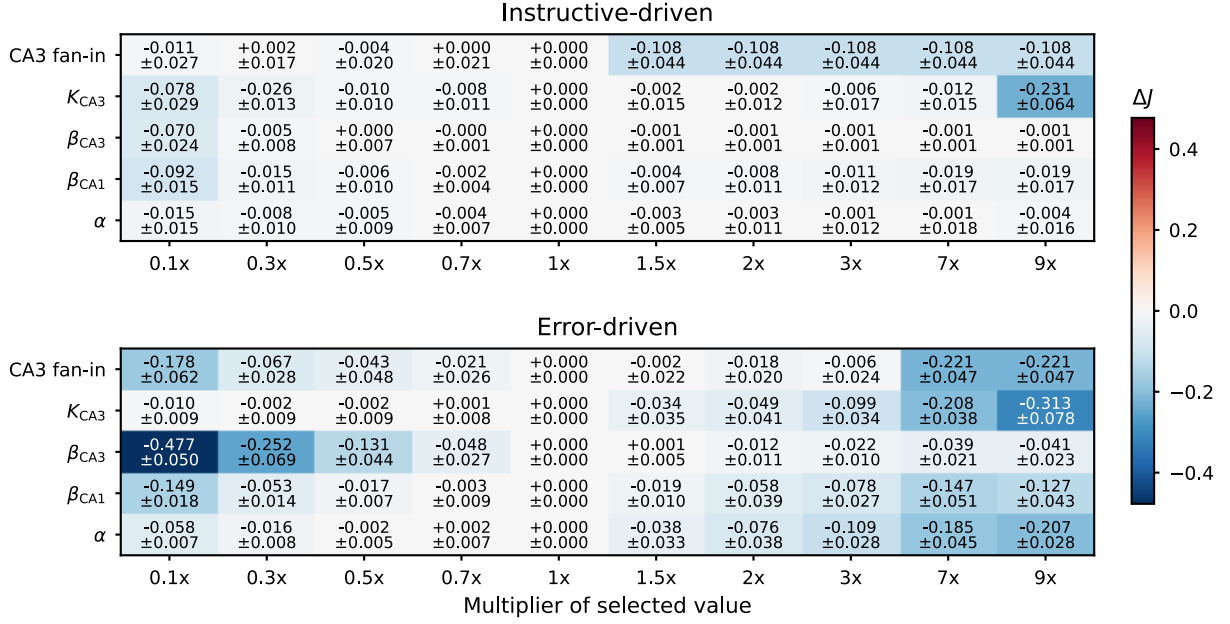
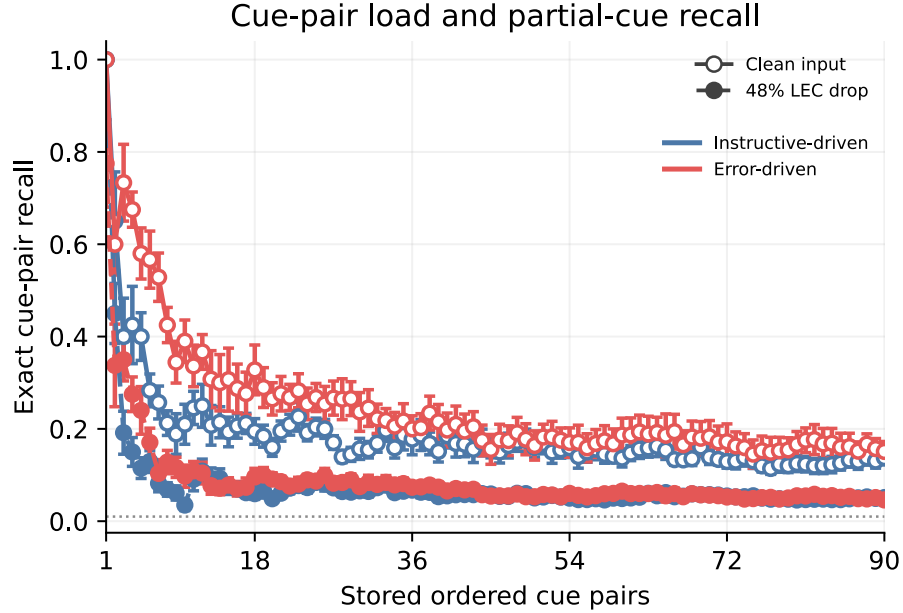
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Figure 4: Parameter sensitivity and cue-pair load in separately optimized operating regimes. (A) Change in the composite recall score when one parameter is multiplied by the indicated factor and the others remain fixed at that rule’s selected values. Integer parameters were rounded and model-valid bounds enforced. Each cell reports the paired mean score change and its across-seed standard deviation over 10 evaluation seeds. (B) Exact ordered-pair recall as load increases from 1 to 90 pairs. Blue denotes the instructive-driven rule and red the error-driven rule. Open solid curves use intact inputs; filled dashed curves use 12-of-25 (48%) LEC dropout. Points show mean $\pm$ SEM across 10 held-out seeds; the dotted line marks nominal chance (1/100). Both panels use rule-specific parameter vectors and therefore compare operating regimes, not update equations alone.

3 Discussion

This study isolates a constraint on rapid associative learning: information stored in a population state must remain accessible to the circuit that reads it. A fixed permutation preserved the values of the instructed CA1 vector but impaired reconstruction by an unchanged decoder. Transforming the decoder with the same permutation restored performance. This rescue follows from the model’s coordinate symmetry, rather than providing an independent empirical discovery. Its value is to separate an incompatible readout from degradation of the target representation. The additional track, degradation, and load assays show how associative learning operates within the compatible basis, and where its performance fails.

The contribution is therefore not the introduction of structured hippocampal representations or of a learning role for CA1. Earlier models assign the direct EC–CA1 pathway a role in learning shared statistical structure [22], and scaffold-based theories use pre-existing retrieval states to organize new associations [2]. Our model makes a related but narrower distinction between retrieval-key construction, CA3–CA1 association, and output decoding. It removes recurrent retrieval dynamics to expose the relation between the latter two operations. This simplification allows a controlled compatibility assay, but precludes claims that the architecture accounts for the broader episodic or spatial capacities of those models.

Pretraining also sets the scope of generalization. In the track assays the autoencoder was exposed to the same generated lap banks later used for associative storage. Novelty therefore resides in the CA3–CA1 associations, not in learning an unseen input distribution. The circuit additionally omits dentate gyrus, explicit inhibition, and consolidation; these omissions make it an assay of the proposed constraint rather than a comprehensive hippocampal model.

Compatibility also differs from representational invariance. A code need not remain unchanged if its variations preserve the output required by a task, or if downstream weights adapt appropriately. Motor-cortical studies demonstrate constraints imposed by an existing activity repertoire [6, 21]; stable latent dynamics further distinguish population-level structure from fixed decoding of individual recorded neurons [5]. Models of self-healing readouts offer a complementary solution in which downstream plasticity tracks changes in the input code [20]. Here the decoder is deliberately frozen. Accordingly, the simulations establish the cost of a mismatch under a fixed readout, not the necessity of preserving CA1 coordinates when recipient circuits are free to adapt. Long-term stability and representational drift remain outside the tested regime.

The two plasticity rules implement different computational uses of an instruction. The instructive-driven rule updates an association without comparing the target with current recall; the error-driven rule corrects under- and overexpressed CA1 components through bounded synaptic changes. Both yielded similar aligned one-shot reconstruction under the reference configuration, but repeated error-driven learning produced higher clean position accuracy and greater cross-context spatial stability. These results show that matching the target basis does not guarantee identical learning dynamics or equally discriminable outputs. Although motivated by rapid instructed plasticity [1, 17, 25], neither rule implements a temporal eligibility trace or a dendritic plateau. In particular, the signed target–recall discrepancy in the error-driven rule is a computational assumption, not an experimentally identified CA1 error signal. Evidence for entorhinal teaching signals and nonspatial BTSP-like field formation supports the motif, but not this exact implementation [4, 7].

The cue-track results reveal reproducible context dependence rather than unconstrained creation of a new map at each transition. Population maps were similar within a cue assignment and

when that assignment recurred. Consecutive-lap tuning similarity fell at exchanges and recovered on the following lap, while individual units showed continuous mixtures of spatial stability and cue modulation. These features are qualitatively related to cue-sensitive changes in hippocampal firing [12, 18] and to learning-related CA1 field reorganization [7, 14, 24]. However, the transition assay changes both the training context and the context used for the recall probe. The boundary discontinuity therefore combines changes in the learned weights with the immediate effect of presenting a different retrieval key. Pausing plasticity during each probe prevents further learning during measurement; it does not isolate the contribution of plasticity to the transition. Fixed-context probes across training, together with context switches at fixed weights, would be needed to distinguish learning-driven reorganization from cue-dependent re-expression. Nor were the simulated unit distributions fitted to biological field statistics or used to classify rate and global remapping.

The degradation experiments likewise support a specific interpretation. Partial input loss left useful output information under the normal key map, and corruption of one input partition preferentially affected its associated readout. Yet there is no recurrent CA3 network or attractor settling: robustness arises from a distributed feedforward input-to-key transformation and the learned key-to-content mapping. Degraded-input reconstruction alone cannot identify a recurrent completion mechanism [13, 19]. The dense common-key control makes every CA3-like unit receive the same EC average and eliminates item-specific keys. Its failure establishes the importance of distinguishable keys in this circuit, not a unique advantage of sparse connectivity. Sparsity-matched and dimensionality-matched controls would be required for that stronger claim.

The constructed MEC/LEC partition also limits anatomical interpretation. Each input half has a designated information type and corresponding readout, so part of the observed dissociation is expected by design. Biological medial and lateral EC contain mixed signals, and both can contribute to nonspatial CA1 learning [3, 4, 10]. Moreover, the retrieval query and reconstruction target are derived from the same EC pattern. Removing input units tests recovery of missing components of a previously experienced pattern; it does not test retrieval of an independently specified episode feature absent from the query. Such a heteroassociative task would provide a stronger test of cue-to-content retrieval.

The optimization-informed analyses extend, but do not replace, the matched rule comparison. Separate searches selected markedly different fan-in, sparsity, gain, and learning-rate combinations. Their sensitivity and load curves therefore compare complete operating regimes, not update equations in isolation. Held-out perturbations indicate that useful performance extends beyond one exact parameter vector, while large changes can be detrimental. One-at-a-time perturbations do not establish global robustness or identify unique biological parameters. Similarly, the 90-pair load assay measures an empirical interference profile. The autoencoder already represents the full cue vocabulary, total training increases with load, and the 10-cue templates are sparser than those in the two-cue assay. Exact-pair reconstruction should not be interpreted as evidence for learning temporal associations between the cues or as a universal storage-capacity estimate.

Several tests follow directly from these limitations. First, varying target coordinates and decoder adaptation independently could compare preservation of a fixed readout with compensatory downstream learning. Second, crossing probe context with learning history would separate immediate key-dependent responses from lasting changes in CA1 weights. Third, load experiments that hold total writes, cue coverage, and template sparsity fixed would clarify the sources of interference. More broadly, biological tests should compare the behavioral consequences of CA1 changes with their effects on recipient populations, rather than treating the magnitude of CA1 change alone as a

measure of memory disruption.

Within these bounds, the model provides a compact demonstration of how a previously established representational basis can support rapid associative storage. Its central implication is relational: the usefulness of a retrieved CA1 state depends on the mapping that interprets it. Readout compatibility is thus a testable constraint on models of hippocampal learning, while its biological implementation and long-term maintenance remain open questions.

4 Methods

4.1 Implementation, reproducibility, and statistical treatment

Simulations were implemented in Python using NumPy and PyTorch and run on CPU. Figures 2 and 3A–D used 20 root seeds (51001–51020); sensitivity used 10 independent root seeds (54001–54010), and the cue-pair load experiment used 10 held-out root seeds (53001–53010). Autoencoder and MTL hyperparameter searches used separate fixed development seeds (41001–41003 and 42001–42003, respectively). Conditions and plasticity rules within an experiment were paired by root seed whenever their parameterization permitted a paired comparison. The experiment scripts write a resolved JSON configuration, compressed NumPy arrays, and a tidy CSV source-data table; plotting functions generate the individual panels from those saved arrays. The composed vector figures do not alter the underlying values.

For the evolution-informed analyses, bounded hyperparameter configurations were selected using the covariance matrix adaptation evolution strategy (CMA-ES), a stochastic derivative-free optimizer [8]. A standalone autoencoder search minimized held-out reconstruction error. Independent MTL searches maximized a composite of clean and selectively degraded recall for each plasticity rule while retaining the common reference autoencoder basis; the two searches therefore report related but non-composed engineering analyses. Search data, objectives, bounds, and selected values are reported in Appendix A. The shared reference configuration was retained separately to provide a matched update-rule comparison and was not produced by the rule-specific searches. CMA-ES is an outer-loop engineering procedure, distinct from Adam-based autoencoder pretraining and from the local CA3–CA1 updates studied here. Optimization generations and development seeds were not counted as experimental replicates.

The unit of computational replication was the root seed, not an individual memory, CA1 unit, track position, or corruption mask. Values were first averaged over observations nested within a seed and then summarized across seeds. For a seed-level statistic y_s , $s = 1, \dots, S$, curves and error bars show $\bar{y} \pm \text{SEM}(y)$, where

$$\bar{y} = \frac{1}{S} \sum_{s=1}^S y_s, \quad \text{SEM}(y) = \sqrt{\frac{1}{S(S-1)} \sum_{s=1}^S (y_s - \bar{y})^2}. \quad (1)$$

Confidence intervals (CIs) reported for the principal 20-seed experiments are two-sided 95% Student- t intervals over seed-level values, $\bar{y} \pm t_{0.975, S-1} \text{SEM}(y)$. Contrasts use paired seed-level differences $d_s = y_s^{(A)} - y_s^{(B)}$ rather than differences between independently aggregated conditions. These intervals quantify sensitivity to sampled initializations and inputs, not uncertainty over a biological population. No null-hypothesis significance tests or multiple-comparison corrections were applied; all analyses are descriptive simulation results.

4.2 EC patterns and the pretrained CA1 basis

EC input, CA1, and CA3-like populations each contained 50 units. A one-hidden-layer autoencoder mapped EC input to a 50-unit CA1 representation and back to a 50-unit EC output. CA1 activity used a differentiable top- K sparse activation with $K_{\text{CA1}} = 5$; output activity used a sigmoid decoder. The shared reference configuration in Figs. 2 and 3 used latent and output gains of 25. The separate autoencoder CMA-ES analysis in Appendix A explored whether those gains could be improved, but its selected values were not substituted into the final MTL searches or figure-generating simulations. Linear encoder and decoder biases were disabled. Specifically, if $z_{(1)} \geq \dots \geq z_{(N)}$ are the sorted preactivations, the activation threshold and sparse nonlinearity were

$$\vartheta_K(z) = \frac{z_{(K)} + z_{(K+1)}}{2}, \quad (2)$$

$$[\mathcal{S}_{K,\beta}(z)]_i = \sigma(\beta[z_i - \vartheta_K(z)]), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \quad (3)$$

Thus approximately K units lay above the adaptive half-activation threshold, while gradients remained defined near that threshold. For EC pattern $x_{\text{EC}} \in \{0, 1\}^{50}$, encoder E , and decoder D , pretraining computed

$$x_{\text{IS}} = \mathcal{S}_{K_{\text{CA1}}, \beta_{\text{latent}}}(Ex_{\text{EC}}), \quad (4)$$

$$\tilde{x}_{\text{EC}} = \sigma(\beta_{\text{out}}Dx_{\text{IS}}), \quad (5)$$

$$\mathcal{L}_{\text{AE}} = \frac{1}{Nd_{\text{EC}}} \sum_{n=1}^N \left\| \tilde{x}_{\text{EC}}^{(n)} - x_{\text{EC}}^{(n)} \right\|_2^2. \quad (6)$$

The weights were trained with Adam (learning rate 0.001, batch size 64) to minimize Eq. 6. Pretraining occurred before CA3–CA1 storage, after which the encoder and decoder were fixed. The encoder therefore supplied the instructed CA1 state (IS), and the decoder defined the output coordinates against which compatibility was tested.

For the readout-compatibility experiment, the autoencoder was trained for 1024 epochs on 2,000 structured MEC+LEC samples and evaluated on 250 held-out samples. Twenty-eight additional patterns were reserved for one-shot associative storage. Samples were drawn from independently generated laps containing both cue assignments and shuffled before training, validation, and storage; the assay therefore preserved marginal MEC+LEC structure without a sequential track traversal. For the cue-reconfiguration experiment, the autoencoder was trained for 256 epochs on the 40 training laps and evaluated on 10 separately generated laps. For the degradation experiment, it was trained for 256 epochs on 20 intact laps and evaluated on five separately generated laps. In the load experiment it was trained for 256 epochs on the complete bank of pair-specific storage laps and evaluated on independently generated pair probes, so the frozen basis did not change with associative load. Each lap contained 50 positions.

Track inputs contained 25 MEC-like and 25 LEC-like units. For position $p \in \{0, \dots, L-1\}$ on a circular track of length $L = 50$, circular distance was $d_L(p, \mu) = \min(|p - \mu|, L - |p - \mu|)$. MEC unit i had a preferred position μ_i from an even tiling of the track and Gaussian activation probability

$$g_i(p) = \exp \left[-\frac{1}{2} \left(\frac{d_L(p, \mu_i)}{\sigma_{\text{MEC}}} \right)^2 \right], \quad \sigma_{\text{MEC}} = 5. \quad (7)$$

Binary MEC activity was sampled independently as $x_{\text{MEC},i}(p) \sim \text{Bernoulli}(g_i(p))$. Two fixed, non-overlapping 25-dimensional LEC cue templates q_1 and q_2 were centered at $c_1 = 10$ and $c_2 = 30$.

Cue j was present with probability

$$g_j^{\text{cue}}(p) = \exp \left[-\frac{1}{2} \left(\frac{d_L(p, c_j)}{4} \right)^2 \right], \quad (8)$$

$$P_j(p) = \text{clip}_{[0,1]} \{ 2\sigma(40[g_j^{\text{cue}}(p) - 0.1]) - 1 \}. \quad (9)$$

If the Bernoulli cue-presence sample was positive, the LEC block was set to q_j ; otherwise it was zero. The complete entorhinal target was the concatenation $x_{\text{EC}}(p) = [x_{\text{MEC}}(p); x_{\text{LEC}}(p)]$. This stochastic construction generated different laps from the same underlying spatial and cue statistics.

Each of the two fixed cue templates activated a disjoint block of $\lfloor 25/2 \rfloor = 12$ LEC units; the remaining unit was unused.

4.3 CA3-like retrieval keys and CA3–CA1 plasticity

EC input was projected to 50 CA3-like units through a fixed sparse matrix followed by a differentiable top- K activation. In the shared reference configuration, all principal experiments used 29 EC inputs per CA3-like unit, $K_{\text{CA3}} = 3$, $\beta_{\text{CA3}} = 170.6$, and $\alpha = 0.092$. The normal track map was constructed to balance EC participation across CA3-like units. CA3–CA1 weights were initialized to zero. The CA1 recall activation used $K_{\text{CA1}} = 5$, with inverse temperature 41.8. Figure 4A,B instead used the rule-specific values selected by CMA-ES and listed in Appendix A.

Writing and recall used the same fixed EC–CA3 projection A and frozen decoder D . At weight state W_t , the complete forward computation was

$$x_{\text{CA3}} = \mathcal{S}_{K_{\text{CA3}}, \beta_{\text{CA3}}}(Ax_{\text{EC}}), \quad (10)$$

$$x_{\text{CA1}} = \mathcal{S}_{5, \beta_{\text{CA1}}}(W_t x_{\text{CA3}}), \quad (11)$$

$$\hat{x}_{\text{EC}} = \sigma(\beta_{\text{out}} D x_{\text{CA1}}). \quad (12)$$

Each selected input to a CA3-like unit had projection weight $1/50$; all other entries of A were zero. Consequently, the convergence parameter controlled which EC components contributed to a key, whereas the top- K nonlinearity controlled key population sparsity. Equation 12 was also used to compute the pre-update recall state required by the error-driven rule.

Let x_{EC} denote the EC input/target, x_{CA3} its retrieval key, x_{IS} the instructed CA1 state produced by the fixed encoder, and x_{CA1} the state recalled through the current CA3–CA1 weights W_t . The instructive-driven update was

$$W_{t+1} = (1 - \alpha x_{\text{IS}}) \odot W_t + \alpha x_{\text{IS}} x_{\text{CA3}}^{\top}, \quad (13)$$

where the first term is broadcast across presynaptic dimensions. For the bounded error-driven rule,

$$e^+ = [x_{\text{IS}} - x_{\text{CA1}}]_+, \quad e^- = [x_{\text{CA1}} - x_{\text{IS}}]_+, \quad (14)$$

$$\delta^+ = \alpha(e^+ x_{\text{CA3}}^{\top})(1 - W_t), \quad \delta^- = \alpha(e^- x_{\text{CA3}}^{\top})W_t, \quad (15)$$

$$W_{t+1} = \text{clip}_{[0,1]}(W_t + \delta^+ - \delta^-). \quad (16)$$

Recalled CA1 activity was obtained by applying the same target sparsity to $W_t x_{\text{CA3}}$ and was passed through the fixed pretrained decoder. The instructive-driven rule depends on the instructed target and presynaptic key but not the current recall error. The error-driven rule separates the residual

into positive and negative components, such that under- and overexpressed CA1 components drive bounded potentiation and depression, respectively. All products between equally shaped arrays were elementwise unless an outer product or matrix product is shown explicitly. Memories were presented sequentially, and W_{t+1} was carried to the next item or position without replay, batching, or gradient descent through the local plasticity dynamics.

4.4 Paired plasticity-rule design

Both rules were evaluated in every main experiment. In the matched reference experiments (Figs. 2 and 3A–D), they shared EC inputs, pretrained autoencoder weights, memory hyperparameters, CA3 wiring seeds, storage order, and—in the degradation experiment—corruption masks. Each rule had its own CA3–CA1 weight matrix, which changed according to the corresponding update. These comparisons therefore isolate the update equation within one common operating regime. Supplementary Fig. S1 collects four paired contrasts derived directly from those result arrays; it is not a separate simulation.

The optimization, sensitivity, and cue-pair load analyses answer a different question: how each rule behaves near a separately selected operating point. Those analyses shared data-generation procedures and random-seed streams but used rule-specific CA3 fan-in, sparsity, gains, and learning rates. Their between-rule differences therefore reflect complete operating regimes and must not be attributed to the update equation alone.

4.5 Readout-compatibility experiment

For each seed, the 28 held-out EC patterns were assigned CA3-like retrieval keys and pretrained CA1 instructions, then written once in a shared random order. The aligned condition used the original IS and decoder. The fixed-permutation condition applied one seed-specific CA1 permutation to every IS while leaving the decoder unchanged. The matched-decoder condition used the same permuted instructions and correspondingly permuted decoder columns. The random-content condition associated each key with a shuffled memory instruction. The no-plasticity condition retained zero CA3–CA1 weights. Recall was evaluated using cosine similarity between reconstructed EC output and the corresponding EC target. Each seed’s plotted value is its mean across 28 memories.

More explicitly, let P be the permutation matrix and $h = x_{\text{IS}}$. The aligned condition stored h and read it with D ; the fixed-permutation condition stored Ph but retained D ; and the matched-decoder condition used Ph together with $D' = DP^T$. In the idealized exact-recall case,

$$D'Ph = DP^TPh = Dh, \quad (17)$$

so matched decoder relabeling restores the original output coordinates without restoring the original CA1 unit identities. The same relation holds throughout learning: starting from zero weights, both update rules and the CA1 activation commute with row permutation, giving $W'_t = PW_t$ and $x'_{\text{CA1}} = Px_{\text{CA1}}$ at each step, up to numerical precision. The matched decoder therefore restores the output even when recall is imperfect. This control isolates coordinate compatibility from a generic effect of permutation. Similarity between two nonzero vectors u and v was

$$\cos(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}; \quad (18)$$

zero-norm cases were assigned similarity zero by the numerical implementation.

4.6 Cue-reconfiguration experiment

Models learned 40 complete laps. Cue identities exchanged positions every 10 laps, alternating between assignments $[0, 1]$ and $[1, 0]$. The matched no-swap schedule retained $[0, 1]$ for all 40 laps. Within a seed, the schedules shared the pretrained autoencoder, initial CA3–CA1 weights, MEC-like trajectory, lap count, plasticity parameters, and the exact EC–CA3 projection; only cue identity after the first 10 laps differed by design. Both plasticity rules used this same projection. Thus all schedule contrasts in Fig. 2B,E,F are paired at the level of the root seed and fixed retrieval-key wiring.

Panels B–D and the event-aligned/population-map analyses in panels E–F were generated by separate scripts using the same root-seed range and protocol but different random substreams for tracks, autoencoders, and probes. Their boundary estimates are therefore separate realizations, not alternative summaries of identical trajectories. Pairing of exchange and no-swap conditions applies within each realization.

After each training lap, plasticity was paused for probing. Panels B–D probed both fixed cue contexts and retained the scheduled-context probe for the transition analysis; panels E–F probed the scheduled context directly. For each CA1 unit, activity was mean-centered across positions before calculating cosine similarity between consecutive spatial tuning curves. Similarities were then averaged across units. Cue-exchange transitions were compared with the same lap boundaries in the paired no-swap model.

Let $f_{\ell ci}(p)$ be the frozen-probe activity of CA1 unit i at position p , after training lap ℓ , under cue context c , and define the centered curve $\tilde{f}_{\ell ci}(p) = f_{\ell ci}(p) - L^{-1} \sum_{r=0}^{L-1} f_{\ell ci}(r)$. Consecutive-lap tuning similarity was

$$T_\ell = \frac{1}{N_{\text{CA1}}} \sum_{i=1}^{N_{\text{CA1}}} \cos(\tilde{f}_{\ell c_\ell i}, \tilde{f}_{\ell-1, c_{\ell-1}, i}). \quad (19)$$

The transition statistic compared T_ℓ at boundaries $\ell \in \{10, 20, 30\}$ with the same boundaries in the matched no-swap run.

For the exact-wiring event analysis, each boundary $e \in \{10, 20, 30\}$ was treated as relative lap $\tau = 0$, the first lap learned under the exchanged cue assignment. The analysis window was $\tau \in \{-3, \dots, 9\}$. Let $c_{e+\tau}$ be the context scheduled at that relative lap. The final pre-exchange probe and the final probe of the ensuing ten-lap block defined the old and adapted references, respectively:

$$R_{\text{old}}(e, \tau) = \frac{1}{N_{\text{CA1}}} \sum_i \cos(\tilde{f}_{e+\tau, c_{e+\tau}, i}, \tilde{f}_{e-1, c_{e-1}, i}), \quad (20)$$

$$R_{\text{adapt}}(e, \tau) = \frac{1}{N_{\text{CA1}}} \sum_i \cos(\tilde{f}_{e+\tau, c_{e+\tau}, i}, \tilde{f}_{e+9, c_{e+9}, i}). \quad (21)$$

The adapted reference is retrospective: it describes convergence toward the state reached at the end of a block and is not a prediction of a future map. Consecutive-lap similarity used Eq. 19 at each $e + \tau$. Values were first averaged over the three events within a root seed and then summarized across seeds. The no-swap control was indexed by the same nominal boundaries, although its cue assignment did not change.

For the population representational-similarity matrix, the complete position-by-unit probe map after lap ℓ was centered independently for each unit over position and vectorized as

$$v_\ell = \text{vec} \left[f_\ell - \frac{1}{L} \mathbf{1} \mathbf{1}^\top f_\ell \right], \quad P_{\ell m} = \cos(v_\ell, v_m). \quad (22)$$

The displayed matrix is the elementwise mean of P across 20 root seeds. Unit-level endpoint similarity was $U_{ei} = \cos(\tilde{f}_{e-1, c_{e-1}, i}, \tilde{f}_{e+9, c_{e+9}, i})$. Its empirical cumulative distribution was computed within each seed after pooling that seed’s three exchange events; the plotted curve and uncertainty were then calculated across seeds, avoiding treatment of individual CA1 units as independent replicates.

After the final lap, plasticity remained frozen and the two cue contexts were probed. For each CA1 unit, spatial stability was the cosine similarity between its two mean-centered context tuning curves. Cue modulation was the mean absolute activity difference between contexts. These metrics quantify context sensitivity but do not themselves classify rate or global remapping. Heat maps show the error-driven seed whose mean stability and modulation were closest to the across-seed medians; units were ordered by preferred position across the two contexts. For final-lap contexts a and b , the plotted unit-level quantities were

$$S_i = \cos(\tilde{f}_{ai}, \tilde{f}_{bi}), \quad (23)$$

$$M_i = \frac{1}{L} \sum_{p=0}^{L-1} |f_{ai}(p) - f_{bi}(p)|. \quad (24)$$

Here S_i measures preservation of spatial profile after removing mean firing, whereas M_i retains absolute context-dependent rate differences; their joint distribution therefore separates two complementary aspects of mixed tuning.

4.7 Selective degradation experiments

Models stored 20 intact laps with cue assignment $[0, 1]$. During frozen recall, a fixed random subset of eligible EC units was zeroed throughout a complete probe lap. Masks were sampled independently between probes. Dropped fractions were 0, 0.25, 0.50, 0.75, 0.90, and 1.00, with 12 masks per fraction and root seed. Because a modality contained 25 units and the implementation rounded $25q$ to an integer, the corresponding numbers removed were 0, 6, 12, 19, 22, and 25 units (effective fractions 0, 0.24, 0.48, 0.76, 0.88, and 1.00). The Results report the 22-of-25 (0.88) partial-degradation point; figure abscissae retain the requested fractions, and the curves continue to complete ablation for reference. LEC-like degradation operated on the final 25 input units; MEC-like degradation operated on the first 25.

For dropped fraction q and mask replicate r , a binary mask $m^{(q,r)} \in \{0, 1\}^{50}$ set $\text{round}(25q)$ randomly selected coordinates in the targeted input half to zero and left all other coordinates at one. The same mask was held fixed for all positions of that probe lap:

$$x_{\text{EC}}^{(q,r)}(p) = m^{(q,r)} \odot x_{\text{EC}}(p). \quad (25)$$

Plasticity was disabled during these probes, so degradation altered retrieval evidence but did not rewrite the stored association.

Cue identity was decoded at the two cue positions by cosine comparison between recalled LEC output and the two fixed cue templates. Spatial accuracy was the fraction of recalled MEC outputs whose nearest clean MEC target had the correct track-position index; chance was therefore $1/50$. MEC output cosine was averaged across all positions. For every curve, the 12 mask-level values were first averaged within a seed before means and uncertainty were calculated across seeds.

Writing $\hat{x}_{\text{LEC}}(c_j)$ for recalled LEC output at cue center c_j and $x_{\text{MEC}}^*(p)$ for the clean MEC target at position p , the three reported metrics were

$$A_{\text{cue}} = \frac{1}{2} \sum_{j=1}^2 \mathbf{1} \left[j = \arg \max_{k \in \{1,2\}} \cos(\hat{x}_{\text{LEC}}(c_j), q_k) \right], \quad (26)$$

$$A_{\text{pos}} = \frac{1}{L} \sum_{p=0}^{L-1} \mathbf{1} \left[p = \arg \max_{u \in \{0, \dots, L-1\}} \cos(\hat{x}_{\text{MEC}}(p), x_{\text{MEC}}^*(u)) \right], \quad (27)$$

$$C_{\text{MEC}} = \frac{1}{L} \sum_{p=0}^{L-1} \cos(\hat{x}_{\text{MEC}}(p), x_{\text{MEC}}^*(p)). \quad (28)$$

Cue accuracy has two-class chance level $1/2$, whereas exact-position accuracy has chance level $1/L = 1/50$. MEC cosine is graded and has no corresponding uniform-classification chance line.

The dense common-key control replaced the EC-CA3 map with a matrix whose every entry was $1/50$. All CA3-like units consequently received the same average EC input and generated a common, non-item-specific key. Because this control also changes connection density and sparsity relative to the normal map, it tests the need for key discriminability rather than the isolated effect of sparse connectivity. In matrix notation the control used $A_{ij} = 1/50$ for all i, j ; hence every CA3 preactivation was identical for a given input. Equation 3 therefore returned the same 0.5-valued vector for every input, rather than an item-specific key.

4.8 Evolution-informed sensitivity and cue-pair load

The local sensitivity analysis started from each rule’s independently selected parameter vector θ_r^* . For parameter j and multiplier $m \in \{0.1, 0.3, 0.5, 0.7, 1, 1.5, 2, 3, 7, 9\}$, it evaluated

$$\Delta J_{rjm} = J_r(\theta_{r,-j}^*, m\theta_{r,j}^*) - J_r(\theta_r^*), \quad (29)$$

using the same composite objective on 10 independent seeds (54001–54010), disjoint from the MTL search seeds. Cells report the mean paired change from the unperturbed vector together with the across-seed standard deviation. Fan-in and K_{CA3} were rounded to integers. Values outside model-valid ranges were clipped, so repeated cells can occur when several multipliers map to the same integer or bound. This is a one-at-a-time local analysis; it does not measure compensatory interactions among parameters or global identifiability.

For the load experiment, $N_{\text{cue}} = 10$ fixed LEC templates generated $N_{\text{cue}}(N_{\text{cue}} - 1) = 90$ ordered, non-repeating assignments to the two cue positions. Within each of 10 held-out seeds, assignments were organized in balanced offset rounds and shuffled within rounds. A capacity condition selected the first $M \in \{1, \dots, 90\}$ assignments from this same bank. The autoencoder basis was pretrained once per seed on laps spanning all 90 assignments and then held fixed across M . Each selected pair was written in three cycles, with pair order independently shuffled within each cycle.

The template generator assigned each of the 10 cues a disjoint block of $\lfloor 25/10 \rfloor = 2$ active LEC units, leaving five units unused. Consequently, this assay differs from the two-cue experiments in both vocabulary size and template sparsity.

Recall used an independently generated intact probe for every stored pair. Cue identity at each location was decoded by maximum cosine similarity to the 10 templates. Exact pair accuracy required both identities to be correct,

$$A_{\text{pair}}(M) = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[(\hat{a}_i, \hat{b}_i) = (a_i, b_i)]. \quad (30)$$

The degraded condition independently requested removal of 50% of LEC units; integer rounding removed 12 of 25 units (48%) with four masks per stored pair. Mask-level exact scores were averaged within pair and then within seed. Independent decoding at two locations gives nominal chance $1/N_{\text{cue}}^2 = 0.01$. Because the encoding basis was exposed to the complete cue vocabulary and the number of training experiences grows with M , this experiment is an operational load curve, not an estimate of an architecture-independent storage capacity. The nominal chance line assumes independent uniform guesses; a decoder restricted to the 90 non-repeating ordered pairs would instead have chance accuracy $1/90$.

Data and code availability

The simulations underlying Figs. 2, 3, 4, and Supplementary Figs. S1 and S2 are generated by the scripts in `src/experiments`. This includes the autoencoder and MTL CMA-ES searches, local parameter-sensitivity analysis, and cue-pair load experiment. Resolved configurations, selected parameter files, compressed numerical arrays, and tidy source-data tables are stored in `results/preprint`. Manuscript Figures 3 and 4 are assembled from saved panels and sensitivity arrays by `compose_preprint_figures.py`; composition does not rerun simulations. A permanent public archive and accession identifier will be added before public posting.

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Supplementary material

Appendix A: Evolutionary hyperparameter selection

Optimization strategy

Hyperparameters were selected with the covariance matrix adaptation evolution strategy (CMA-ES), a stochastic, derivative-free optimizer for nonlinear and non-convex objectives [8]. CMA-ES sampled a multivariate normal population in standardized coordinates and adapted its mean, covariance, and global step size from candidate rankings. If g is a standardized genome, model parameters were decoded as

$$\theta(g) = \text{clip}_{[\ell, u]}(c + s \odot g), \quad (31)$$

where c , s , ℓ , and u are the parameter-wise centers, scales, lower bounds, and upper bounds. Count-valued parameters were rounded to the nearest integer after clipping. A quadratic penalty discouraged samples outside the standardized bounds before clipping. Candidate evaluations were parallelized across CPU processes, but parallelization did not change the objective.

The search was used only for engineering model selection. Model weights were not encoded in the genome: autoencoder weights were still learned with Adam, and CA3–CA1 weights were still changed by the local plasticity rule. Fixed development seeds made candidate evaluations deterministic and comparable. No seed was optimized separately, and development seeds were disjoint from the 20 seeds used for the matched principal experiments, the 10 independent seeds used for sensitivity analysis, and the 10 seeds used for the cue-pair load test. A single CMA-ES run was performed for each reported search, so the results characterize the selected run rather than variability across independent optimizer restarts.

Autoencoder objective

The autoencoder search varied K_{CA1} , the latent sparse-activation gain β_{latent} , and the decoder gain β_{out} . It used three development seeds (41001–41003). For each candidate and seed, separate autoencoders were trained for 64 epochs on (i) 1,024 sparse binary EC patterns and (ii) 20 MEC+LEC track laps, then tested on 128 disjoint sparse patterns and five independently generated track laps. The minimized objective was

$$J_{\text{AE}}(\theta) = \frac{1}{3} \sum_{s=1}^3 \left[\frac{1}{2} \mathcal{L}_{\text{sparse}, s}(\theta) + \frac{1}{2} \mathcal{L}_{\text{track}, s}(\theta) \right], \quad (32)$$

where each $\mathcal{L}$ is mean squared validation reconstruction error. The search used 96 generations of seven candidates (672 candidate evaluations) with boundary-penalty coefficient 0.1. The selected values were $K_{\text{CA1}} = 5$, $\beta_{\text{latent}} = 3.912370$, and $\beta_{\text{out}} = 25.333004$. Their mean sparse, track, and equally weighted validation MSEs were 0.001770, 0.000593, and 0.001182, respectively.

Medial-temporal-lobe (MTL) recall objective

The autoencoder search above documents an alternative basis optimization. For the final MTL searches and all figure-generating simulations, however, we retained the common reference basis

($K_{\text{CA1}} = 5$, $\beta_{\text{latent}} = \beta_{\text{out}} = 25$), avoiding a simultaneous change of representational basis and associative rule. Five MTL parameters were then optimized independently for the instructive-driven and error-driven rules: EC–CA3 fan-in F , CA3 target activity K_{CA3} , CA3 and CA1 inverse-temperature gains, and plasticity rate α . For each of three development seeds (42001–42003), one autoencoder, one bank of clean training and validation laps, and fixed MEC/LEC corruption masks were prepared before optimization and reused for every candidate and both rules. Storage vectors were shuffled within each generated lap, with cue locations carried along with their samples. This preserved the marginal MEC+LEC distribution while removing sequential adjacency and traversal order.

After 20 clean storage laps, three components were calculated. Clean recall was the mean of full-output cosine, exact-position accuracy, and cue-identity accuracy,

$$J_{\text{clean}} = \frac{1}{3}(C_{\text{EC}} + A_{\text{pos}} + A_{\text{cue}}). \quad (33)$$

The requested 85% dropout removed 21 of 25 units (84%) in either modality. At MEC dropout, the score averaged position and cue accuracy; at LEC dropout, it averaged cue accuracy and MEC output cosine:

$$J_{\text{MECdrop}} = \frac{1}{2}(A_{\text{pos}}^{\text{MECdrop}} + A_{\text{cue}}^{\text{MECdrop}}), \quad (34)$$

$$J_{\text{LECdrop}} = \frac{1}{2}(A_{\text{cue}}^{\text{LECdrop}} + C_{\text{MEC}}^{\text{LECdrop}}). \quad (35)$$

Each degraded component was averaged over six fixed masks per modality. CMA-ES maximized the equally weighted mean over components and development seeds,

$$J_{\text{MTL}}(\theta, r) = \frac{1}{3} \sum_{s=1}^3 \frac{J_{\text{clean},s} + J_{\text{MECdrop},s} + J_{\text{LECdrop},s}}{3}, \quad (36)$$

for rule r . Each rule used 96 generations of eight candidates (768 candidate evaluations) and boundary-penalty coefficient 0.05.

Table 1: MTL CMA-ES search domains and selected values. Fan-in and K_{CA3} are integer counts. Scores are evaluated on the three fixed development seeds and are therefore optimization, not holdout, performance.

Parameter	Search range	Center	Instructive-driven	Error-driven
EC–CA3 fan-in F	1–49	29	36	9
K_{CA3}	1–15	3	6	5
β_{CA3}	5–300	170.6	286.747318	54.400011
β_{CA1}	5–120	41.8	9.160055	26.928441
α	0.005–0.400	0.092	0.033657	0.081975
J_{clean}	–	–	0.700864	0.935067
J_{MECdrop}	–	–	0.539444	0.565000
J_{LECdrop}	–	–	0.901139	0.953269
J_{MTL}	–	–	0.713816	0.817779

The two searches converged to quantitatively different retrieval regimes. The instructive-driven solution used broader EC–CA3 fan-in and a sharper CA3 activation but a softer CA1 recall

nonlinearity; the error-driven solution used narrower fan-in, lower CA3 gain, steeper CA1 recall, and a larger learning rate. These values should not be interpreted as biological estimates or as proof of global optimality. They indicate that, under the stated objective and bounds, the two update rules favor different computational regimes. The local multiplier analysis in Fig. 4A probes the immediate neighborhood of these selected vectors; Eq. 29 and its limitations are described in Methods.

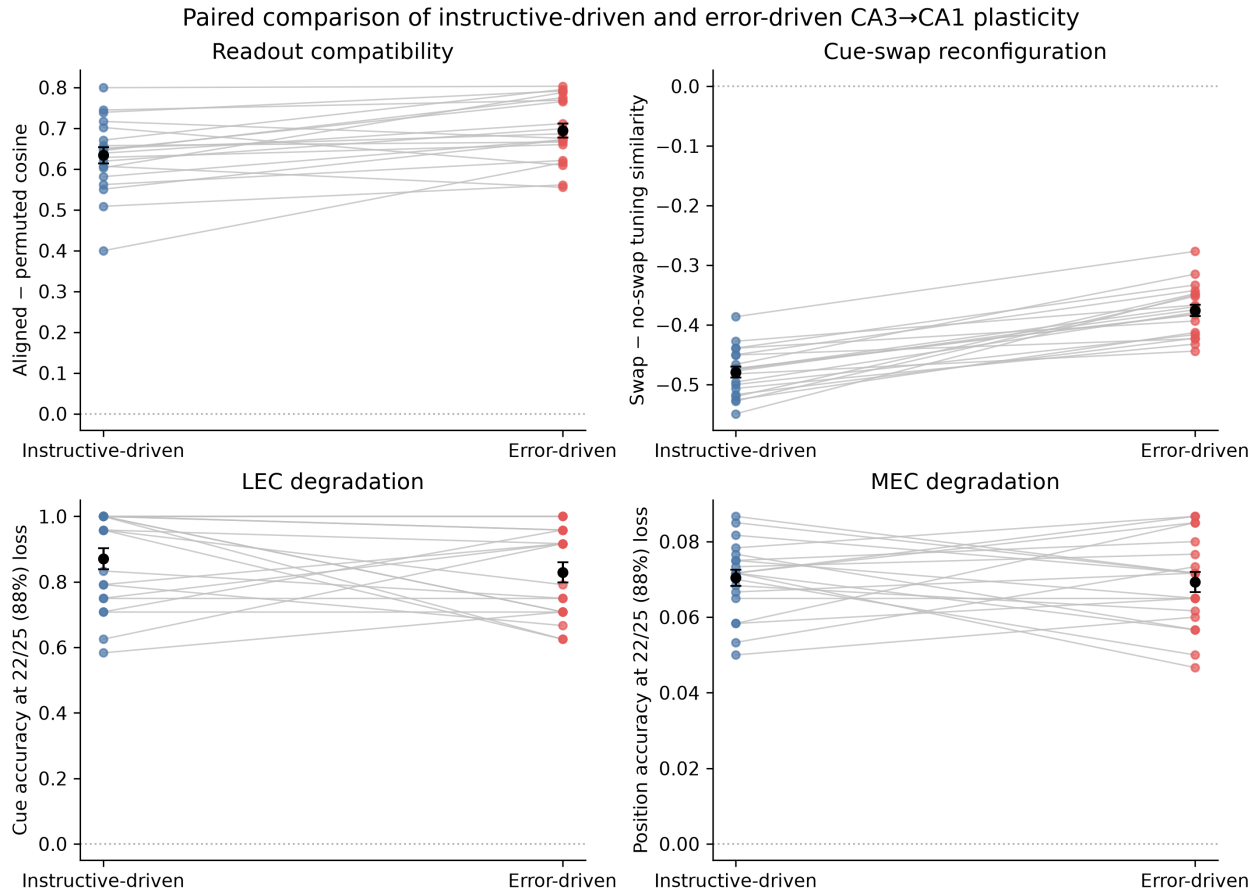


Figure S1: Seed-paired comparison of instructive-driven and error-driven plasticity. Each grey line joins instructive-driven and error-driven values obtained from the same root seed; colored points are individual seed-level values and black points with error bars show mean $\pm$ SEM across 20 seeds. **(Upper left)** Readout-compatibility effect, defined as aligned minus fixed-permutation output-target cosine in Fig. 2A. **(Upper right)** Cue-exchange effect, defined as tuning similarity at exchange boundaries minus similarity at the same boundaries in the no-swap control in Fig. 2B. **(Lower left)** Cue identity accuracy after 22-of-25 (88%) LEC-like degradation in Fig. 3A. **(Lower right)** Nearest-position accuracy after 22-of-25 (88%) MEC-like degradation in Fig. 3C. These contrasts refer to the shared-reference experiments in Figs. 2 and 3A–D: both rules shared inputs, pretrained autoencoders, memory hyperparameters, CA3 wiring seeds, storage order, and corruption masks; only the CA3–CA1 update differed. They are distinct from the rule-specific evolved operating regimes in Fig. 4A,B and Table 1.

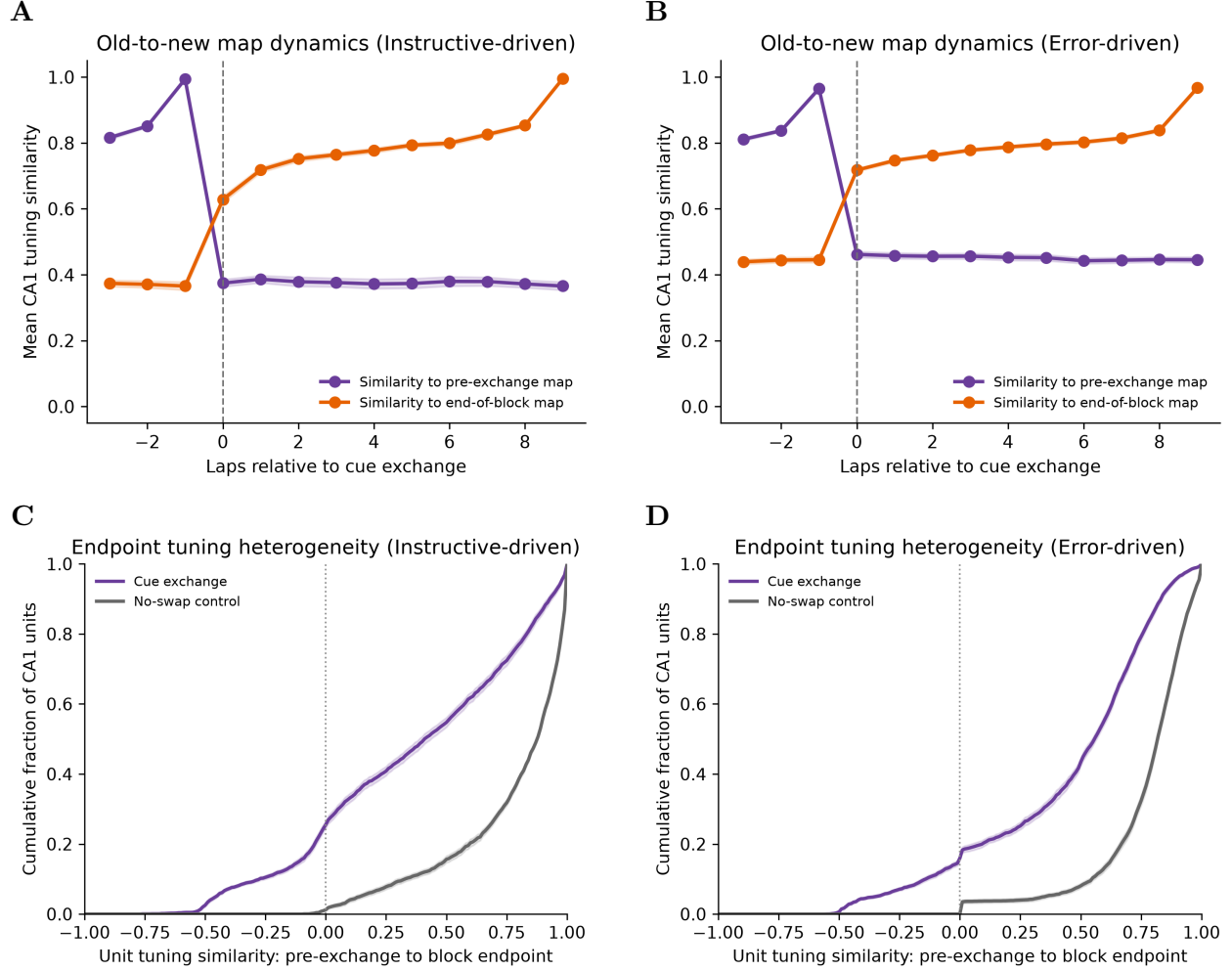


Figure S2: Reference-map dynamics and heterogeneity after cue exchange. (A,B) Similarity of the current CA1 tuning map to the final pre-exchange map (purple) and to the final map of the ensuing ten-lap block (orange), for instructive-driven (A) and error-driven (B) plasticity. Relative lap zero is the first lap learned after cue exchange. The adapted-map reference is retrospective and therefore equals one at the block endpoint by construction; the informative result is the abrupt crossover at exchange followed by smaller within-block changes. (C,D) Seed-averaged empirical cumulative distributions of unit-level tuning similarity between the pre-exchange and block-end maps for instructive-driven (C) and error-driven (D) plasticity. Purple denotes cue exchange and grey the matched no-swap control. For every curve, events and units were aggregated within each root seed before the mean and SEM were calculated across 20 seeds. These distributions describe population heterogeneity and do not treat units as independent replicates. All panels use the exact same EC-CA3 projection in the exchange and no-swap schedules.